

Kinova Gen3 Robot: Rigid Structure and Kinematics

This document explains the rigid body structure and kinematics of the Kinova Gen3 robot using MATLAB. The robot is modeled as a serial manipulator with 7 degrees of freedom (DOF). The following sections describe each link and joint from bottom to top with corresponding visual references.

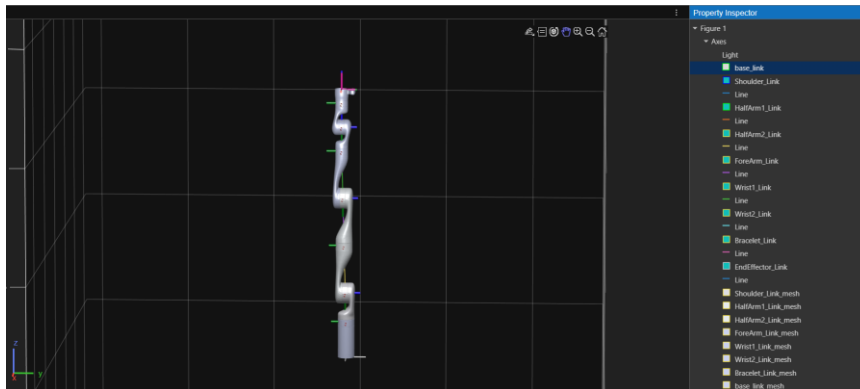
KINOVA GEN 3 :

The **rigidBodyTree** is a representation of the connectivity of rigid bodies with joints. A rigid body tree model is made up of rigid bodies as **rigidBody** objects. Each rigid body has a **rigidBodyJoint** object associated with it that defines how it can move relative to its parent body.

For a given rigid body tree model, you can also use the robot model to calculate joint angles for desired end-effector positions using the robotics inverse kinematics algorithms. Specify your rigid body tree model when using **inverseKinematics** or **generalizedInverseKinematics**.

Robot dynamics calculations are also possible. Specify the **Mass**, **CenterOfMass**, and **Inertia** properties for each **rigidBody** in the robot model. You can calculate forward and inverse dynamics with or without external forces and compute dynamics quantities given robot joint motions and joint inputs. To use the dynamics-related functions, set the `DataFormat` property to "row" or "column".

Index	Body Name	Joint Name	Joint Type	Parent Link
1	Shoulder_Link	Actuator1	Revolute	base_link
2	HalfArm1_Link	Actuator2	Revolute	Shoulder_Link
3	HalfArm2_Link	Actuator3	Revolute	HalfArm1_Link
4	ForeArm_Link	Actuator4	Revolute	HalfArm2_Link
5	Wrist1_Link	Actuator5	Revolute	ForeArm_Link
6	Wrist2_Link	Actuator6	Revolute	Wrist1_Link
7	Bracelet_Link	Actuator7	Revolute	Wrist2_Link
8	EndEffector_Link	Endeffector	Fixed	Bracelet_Link



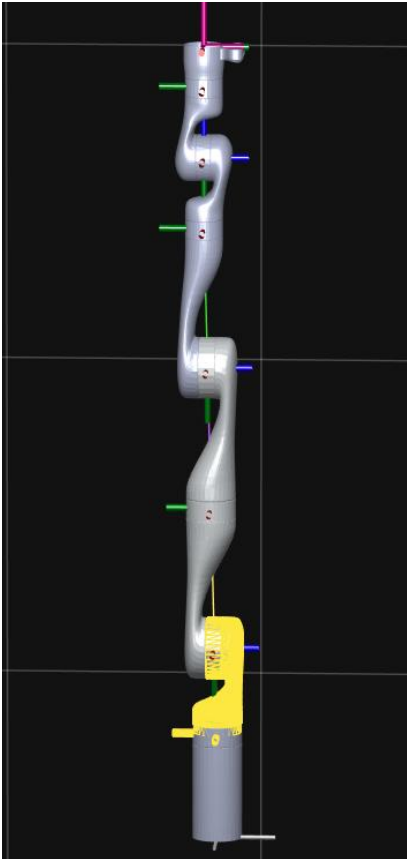
1.

Base and Actuator 1 (Shoulder Link): This is the fixed base of the robot. The first revolute joint allows rotation about the vertical axis.



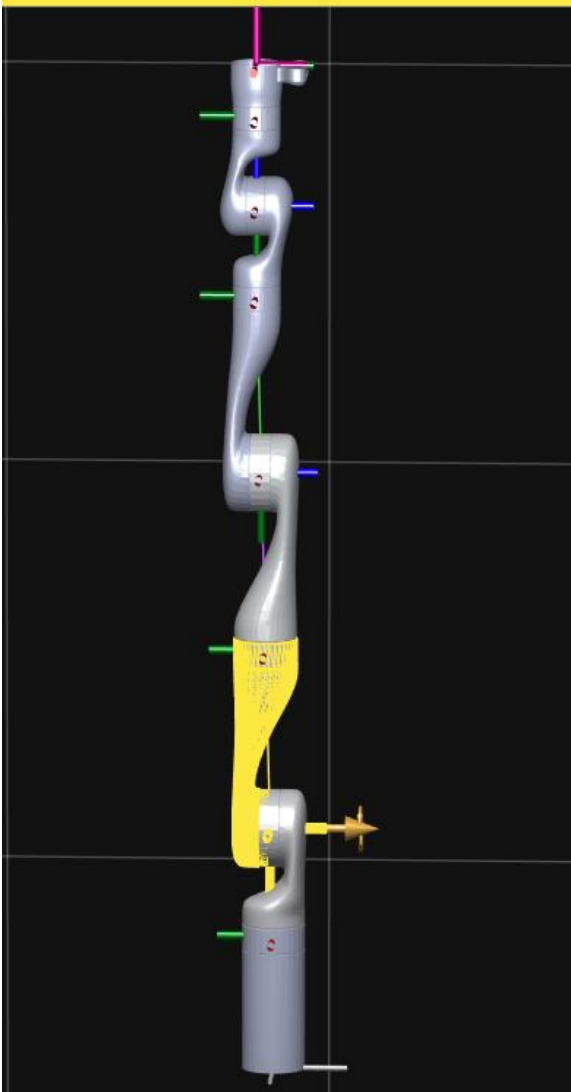
2.

Half Arm 1 (Actuator 2): Provides the first major lifting motion. This joint controls the initial elevation of the arm



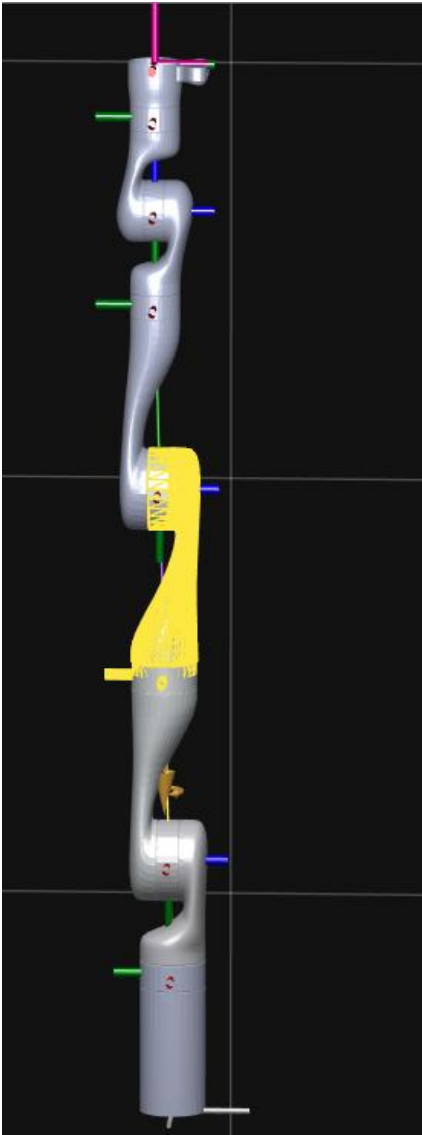
3.

Half Arm 2 (Actuator 3): Extends the arm further and contributes to reach and positioning.



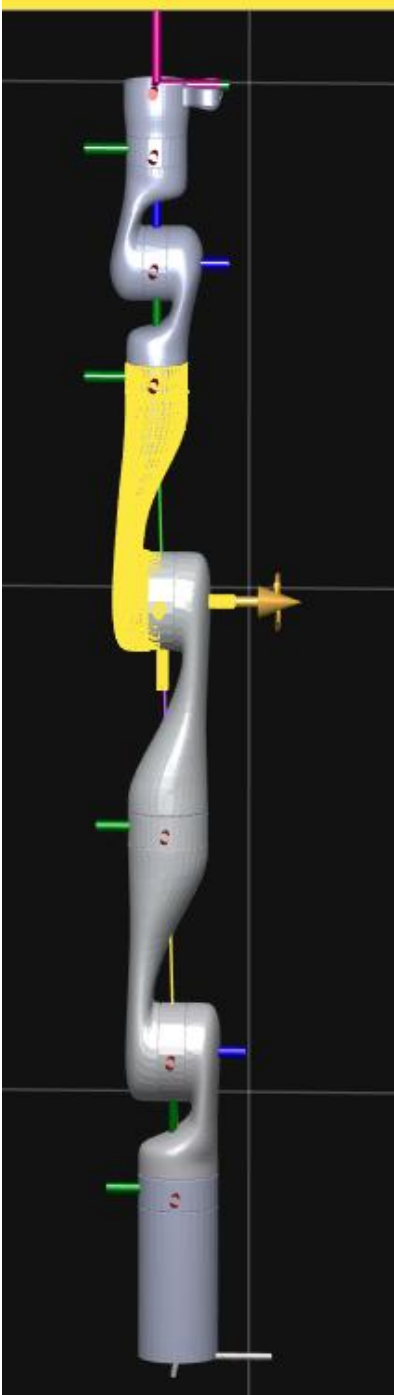
4.

Forearm (Actuator 4): Controls bending of the arm, enabling mid-level positioning in workspace.



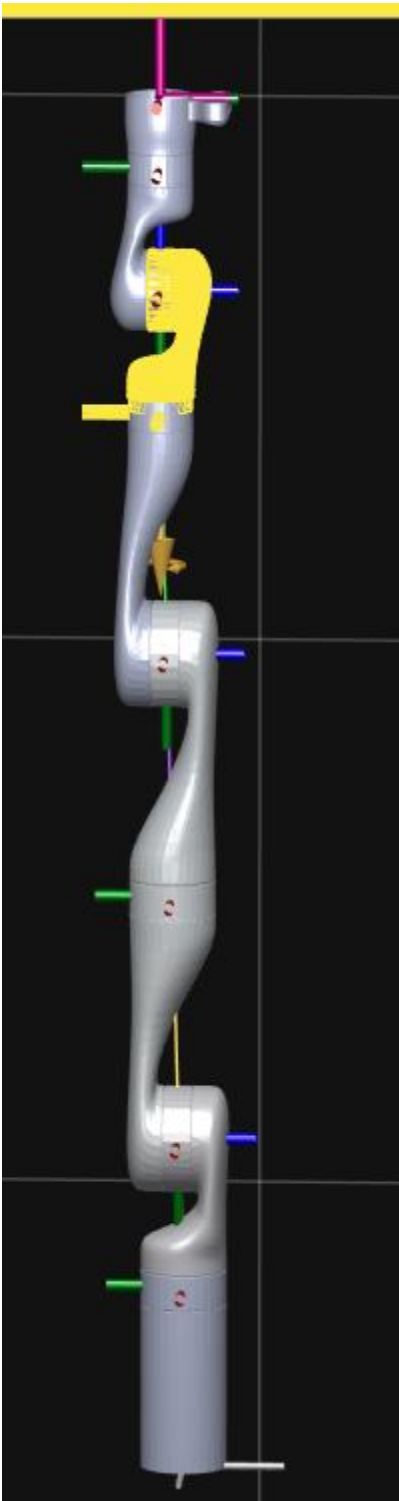
5.

Wrist 1 (Actuator 5): Introduces orientation control for the end-effector.



6.

Wrist 2 (Actuator 6): Adds additional rotational flexibility for precise alignment.



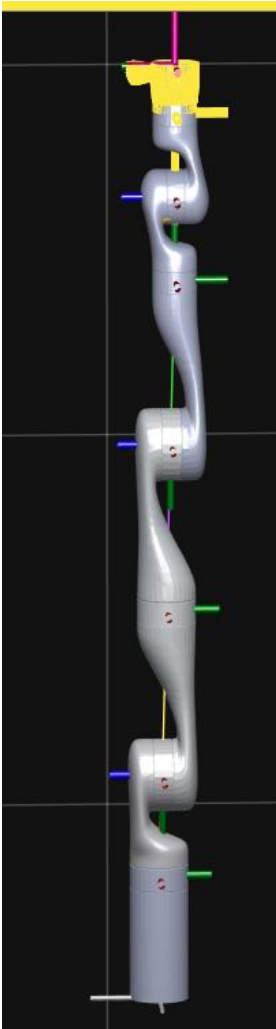
7.

Bracelet Link (Actuator 7): Final rotational joint before the end-effector, enabling fine adjustments.



8.

End Effector: Fixed joint representing the tool frame. Used for interaction with objects.



Data Format- I/O for Kinematics and Dynamics

Input/output data format for kinematics and dynamics functions, specified as "struct", "row", or "column". To use dynamics functions, you must use either "row" or "column".

NEXT STEPS:

Kinematic and Dynamic Analysis of the manipulator with the Sourced End-Effector, YOLOv8
Integration and Importin to Isaac sim.